

12 (a) e.g. logarithm provides a smaller number
gain of amplifiers is series found by addition, (not multiplication)
(*any sensible suggestion*)

B1 [1]

(b) (i) optic fibre

B1 [1]

(ii) attenuation/dB = $10 \lg(P_2/P_1)$
= $10 \lg(\{6.5 \times 10^{-3}\}/\{1.5 \times 10^{-15}\})$
= 126

C1

C1

length = $126 / 1.8$
= 70 km

A1 [3]